FUNCTORIALITY OF SMALL AND BIG ETALE TOPOI: I_F = F_BIG O I_T, F_SMALL WITH PUSHFORWARD, THE COMMUTATIVE SQUARE, AND F_SMALL = PI_X O F_BIG O I_Y

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Editorial note. Parts (1), (3), (4) are argued in-span from the functor descriptions; part (2)'s existence statement is delegated to the properties-of-spaces functoriality lemma (normal cross-chapter citation). The constructs i_f, i_T, pi_X, f_big are defined by same-chapter lemmas whose proofs are not carded; these statement-level uses are declared external, never re-proven.

Allowed standard prerequisites (editorial).

- Morphism of sites from base change. Properties of Spaces, lemma functoriality-etale-site (cross-chapter, named). Part (2): existence of f_spaces,etale and f_small with the pushforward formula.
- Inclusion morphisms i_f, i_T, pi_X, pi_Y and f_big. lemma-put-in-T-etale, lemma-at-the-bottom-etale, lemma-morphism-big-etale (same chapter, NOT carded, declared external). The constructs whose interplay the four parts relate.

Source reference. Original lemma 0DF3: sources/stacks-spaces-topologies/pinned-original-source.tex, source lines 543–543. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0DF2: sources/stacks-spaces-topologies/pinned-original-source.tex, source lines 511–511. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03G2: focused-seeds/geometry/S9-topo-morphism-big-small-etale/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : Y \to X$ be a morphism in $(Spaces/S)_{étale}$.*

(1) *We have $i_f = f_{big} \circ i_T$ with i_f as in Lemma 0DF2 and i_T as in Lemma 0DF3.*

Date: stacks-project:spaces-topologies.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(2) *The functor* $X_{spaces,étale} \to T_{spaces,étale}$, $(U \to X) \mapsto (U \times_X Y \to Y)$ *is continuous and induces a morphism of sites*

$$f_{spaces,étale} : Y_{spaces,étale} \longrightarrow X_{spaces,étale}$$

The corresponding morphism of small étale topoi is denoted

$$f_{small} : Sh(Y_{étale}) \to Sh(X_{étale})$$

We have $f_{small,*}(\mathcal{F})(U/X) = \mathcal{F}(U \times_X Y/Y)$.

(3) *We have a commutative diagram of morphisms of sites*

$$\begin{array}{ccc} Y_{spaces,étale} & \xleftarrow{\pi_Y} & (Spaces/Y)_{étale} \\ \downarrow{\scriptstyle f_{spaces,étale}} & & \downarrow{\scriptstyle f_{big}} \\ X_{spaces,étale} & \xleftarrow{\pi_X} & (Spaces/X)_{étale} \end{array}$$

so that $f_{small} \circ \pi_Y = \pi_X \circ f_{big}$ *as morphisms of topoi.*

(4) *We have* $f_{small} = \pi_X \circ f_{big} \circ i_Y = \pi_X \circ i_f$.

Proof. The equality $i_f = f_{big} \circ i_Y$ follows from the equality $i_f^{-1} = i_T^{-1} \circ f_{big}^{-1}$ which is clear from the descriptions of these functors above. Thus we see (1).

The functor $u : X_{spaces,étale} \to Y_{spaces,étale}$, $u(U \to X) = (U \times_X Y \to Y)$ was shown to give rise to a morphism of sites and correspong morphism of small étale topoi in Properties of Spaces, Lemma 03G2. The description of the pushforward is clear.

Part (3) follows because π_X and π_Y are given by the inclusion functors and $f_{spaces,étale}$ and f_{big} by the base change functors $U \mapsto U \times_X Y$.

Statement (4) follows from (3) by precomposing with i_Y. □